Midterm review

Tuesday, October 21, 2025 4:49 PM

mass : $m = \iint_R \rho \, dx\,dy$

center of mass : $\bar{x} = \frac{1}{m} \iint_R x\rho \, dx\,dy$

$\bar{y} = \frac{1}{m} \iint_R y\rho \, dx\,dy$

cylindrical : $x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$

$$\iiint f(x,y,z)\, r \, dz\,dr\,d\theta$$

spherical : $x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi$

$$\iiint f(x,y,z)\, \rho^2\sin\phi \, d\rho\, d\theta\, d\phi$$

Jacobians : change of variables

$$\iiint_R f(x,y)\,dx\,dy = \iiint_E f(x,y) \, \underset{\uparrow}{J} \, du\,dv$$

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial w} \end{vmatrix} \leftarrow \text{must be absolute value}$$

Surface area : $A = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \, dA$

Line integral : $\int_C f(x,y)\, \underline{ds} = \int_a^b f(x(t), y(t)) \underbrace{\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt}_{ds}$

$\int_C f(x,y)\, \underline{dx} = \int_a^b f(x(t), y(t)) \underbrace{x'(t)\, dt}_{dx}$

Line integral for vector field : $\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)\, dt = \int_C \vec{F} \cdot \vec{T}\, ds$

parametrize C

Mass : $m = \int_C \rho \, ds$

center of mass : $\bar{x} = \frac{1}{m} \int_C x\rho \, ds$

$\bar{y} = \frac{1}{m} \int_C y\rho \, ds$

Fundamental theorem : for a conservative vector field,

$$\int_C \nabla f \cdot dr = f(r(b) - r(a))$$

Conservative : $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, where $\vec{F} = \langle P(x,y), Q(x,y) \rangle$

↙ line integral over closed loop ↔ double integral over region enclosed

Green's theorem : $\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$

parametrize (like r, θ)

Green's theorem, vector form : $\oint_C \vec{F} \cdot d\vec{r} = \iint_D (\nabla \times \vec{F}) \cdot \hat{k} \, dA$

$\operatorname{div} \vec{F} = \nabla \cdot \vec{F}$

$\operatorname{curl} \vec{F} = \nabla \times \vec{F}$

Tangent planes : If curve is parametrized by u, v,

$r_u = \left\langle \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right\rangle$

$r_v = \left\langle \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right\rangle$

$r_u \times r_v \neq 0$ for smooth surfaces

Surface area : $A(S) = \iint_D |r_u \times r_v| \, dA$

$= \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \, dA$

Surface integral : $\iint_S f(x,y,z)\, dS = \iint_D f(r(u,v))\, |r_u \times r_v| \, dA$

$|r_u \times r_v| = \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}$ for $g = z(x,y)$

oriented surface : $\frac{r_u \times r_v}{|r_u \times r_v|}$

flux : $\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \vec{n}\, dS = \iint_D \vec{F} \cdot (\vec{r}_u \times \vec{r}_v)\, dA$

↙ line integral around boundary curve of surface ↔ surface integral over the surface

Stoke's Theorem : $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$

↙ surface integral over closed surface ↔ triple integral over volume enclosed

Divergence Theorem : $\iint_S \vec{F} \cdot d\vec{S} = \iiint_E \nabla \cdot \vec{F} \, dV$